

手摸手实现日志组件库

组件地址

```
<dependency>
  <groupId>org.opengoofy.index12306</groupId>
  <artifactId>index12306-log-spring-boot-starter</artifactId>
  <version>${project.version}</version>
</dependency>
```

组件概述

1. 方法或类上添加日志打印注解并打印日志。

组件功能

日志打印

定义日志打印注解，可以标记在类或者方法上。

标记在类上，类下所有方法都会打印；标记在方法上，仅打印标记方法；如果类或者方法上都有标记，以方法上注解为准。

```
package org.opengoofy.index12306.framework.starter.log.annotation;

import java.lang.annotation.ElementType;
import java.lang.annotation.Retention;
import java.lang.annotation.RetentionPolicy;
import java.lang.annotation.Target;

/**
 * Log 注解打印，可以标记在类或者方法上
 * 标记在类上，类下所有方法都会打印；标记在方法上，仅打印标记方法；如果类或者方法上都有标记，
 * 以方法上注解为准
 */
```

```

* @公众号：马丁玩编程，回复：加群，添加马哥微信（备注：12306）获取项目资料
*/
@Target({ElementType.TYPE, ElementType.METHOD})
@Retention(RetentionPolicy.RUNTIME)
public @interface ILog {

    /**
     * 入参打印
     *
     * @return 打印结果中是否包含入参，{@link Boolean#TRUE} 打印，{@link Boolean#FALSE} 不
    打印
     */
    boolean input() default true;

    /**
     * 出参打印
     *
     * @return 打印结果中是否包含出参，{@link Boolean#TRUE} 打印，{@link Boolean#FALSE} 不
    打印
     */
    boolean output() default true;
}

```

定义日志打印规约参数，共打印参数：开始时间、请求入参、以及返回参数等。

```

import lombok.Data;

/**
 * ILog 日志打印实体
 *
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 */
@Data
public class ILogPrintDTO {

    /**
     * 开始时间
     */
    private String beginTime;

    /**

```

```

    * 请求入参
    */
    private Object[] inputParams;

    /**
    * 返回参数
    */
    private Object outputParams;
}

```

定义日志注解 AOP 扫描类，执行方法执行前后增强，最终打印方法执行日志。

`@within` 和 `@annotation` 注解分别对注解在类上和在方法上增强。

```

package org.opengoofy.index12306.framework.starter.log.core;

import cn.hutool.core.date.DateUtil;
import cn.hutool.core.date.SystemClock;
import com.alibaba.fastjson2.JSON;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import org.aspectj.lang.ProceedingJoinPoint;
import org.aspectj.lang.annotation.Around;
import org.aspectj.lang.annotation.Aspect;
import org.aspectj.lang.reflect.MethodSignature;
import org.opengoofy.index12306.framework.starter.log.annotation.ILog;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.context.request.RequestContextHolder;
import org.springframework.web.context.request.ServletRequestAttributes;
import org.springframework.web.multipart.MultipartFile;

import java.lang.reflect.Method;
import java.util.Optional;

/**
 * {@link ILog} 日志打印 AOP 切面
 *
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 */
@Aspect

```

```

public class ILogPrintAspect {

    /**
     * 打印类或方法上的 {@link ILog}
     */
    @Around("@within(org.opengoofy.index12306.framework.starter.log.annotation.ILog) ||
@annotation(org.opengoofy.index12306.framework.starter.log.annotation.ILog)")
    public Object printMLog(ProceedingJoinPoint joinPoint) throws Throwable {
        long startTime = SystemClock.now();
        MethodSignature methodSignature = (MethodSignature) joinPoint.getSignature();
        Logger log = LoggerFactory.getLogger(methodSignature.getDeclaringType());
        String beginTime = DateUtil.now();
        Object result = null;
        try {
            result = joinPoint.proceed();
        } finally {
            Method targetMethod =
joinPoint.getTarget().getClass().getDeclaredMethod(methodSignature.getName(),
methodSignature.getMethod().getParameterTypes());
            ILog logAnnotation =
Optional.ofNullable(targetMethod.getAnnotation(ILog.class)).orElse(joinPoint.getTarget().
getClass().getAnnotation(ILog.class));
            if (logAnnotation != null) {
                ILogPrintDTO logPrint = new ILogPrintDTO();
                logPrint.setBeginTime(beginTime);
                if (logAnnotation.input()) {
                    logPrint.setInputParams(buildInput(joinPoint));
                }
                if (logAnnotation.output()) {
                    logPrint.setOutputParams(result);
                }
                String methodType = "", requestURI = "";
                try {
                    ServletRequestAttributes servletRequestAttributes =
(ServletRequestAttributes) RequestContextHolder.getRequestAttributes();
                    assert servletRequestAttributes != null;
                    methodType = servletRequestAttributes.getRequest().getMethod();
                    requestURI = servletRequestAttributes.getRequest().getRequestURI();
                } catch (Exception ignored) {
                }
                log.info("[{}] {}", executeTime: {}ms, info: {}, methodType, requestURI,

```

```
SystemClock.now() - startTime, JSON.toJSONString(logPrint));
    }
}
return result;
}

private Object[] buildInput(ProceedingJoinPoint joinPoint) {
    Object[] args = joinPoint.getArgs();
    Object[] printArgs = new Object[args.length];
    for (int i = 0; i < args.length; i++) {
        if ((args[i] instanceof HttpServletRequest) || args[i] instanceof
        HttpServletResponse) {
            continue;
        }
        if (args[i] instanceof byte[]) {
            printArgs[i] = "byte array";
        } else if (args[i] instanceof MultipartFile) {
            printArgs[i] = "file";
        } else {
            printArgs[i] = args[i];
        }
    }
    return printArgs;
}
}
```

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